



JANUS

KEEPING YOUR DRONES SAFE

Team Celeritas :

Anish Jois, Team Lead

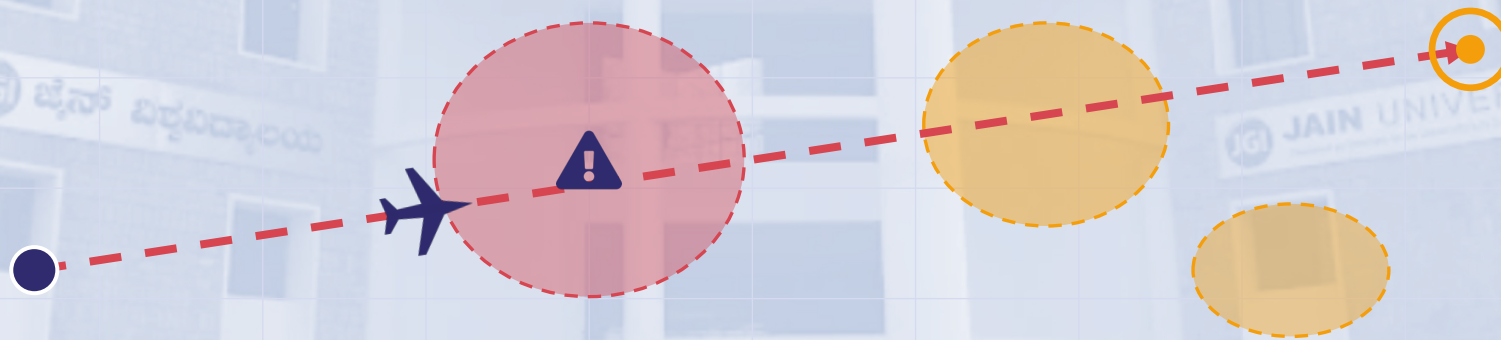
Shlok Tiwari, Programmer

Hemvarshith YVP, Designer

Trivia: *Janus* is the Roman God of pathways and navigation into the unknown

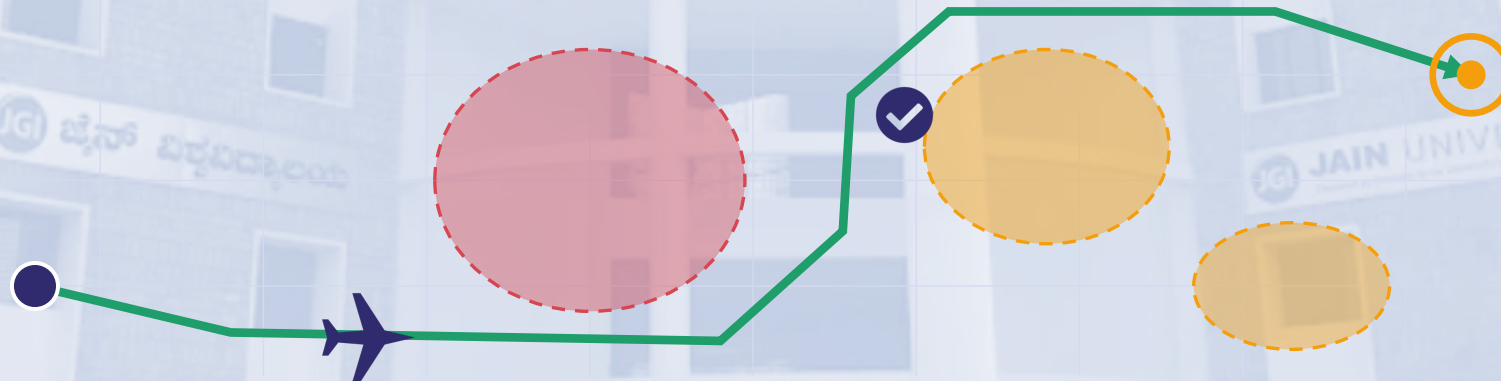
The Problem

Drones and UAVs often fly through Hazardous atmospheric conditions which cause long-term damage



Our solution

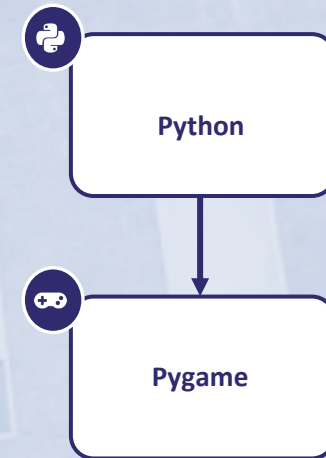
An automatic flight-planner which allows for real time adjustments based on varying atmospheric or flight profiles in a cost-effective manner



Tech Stack

I. Front-end / UI : Python + Pygame

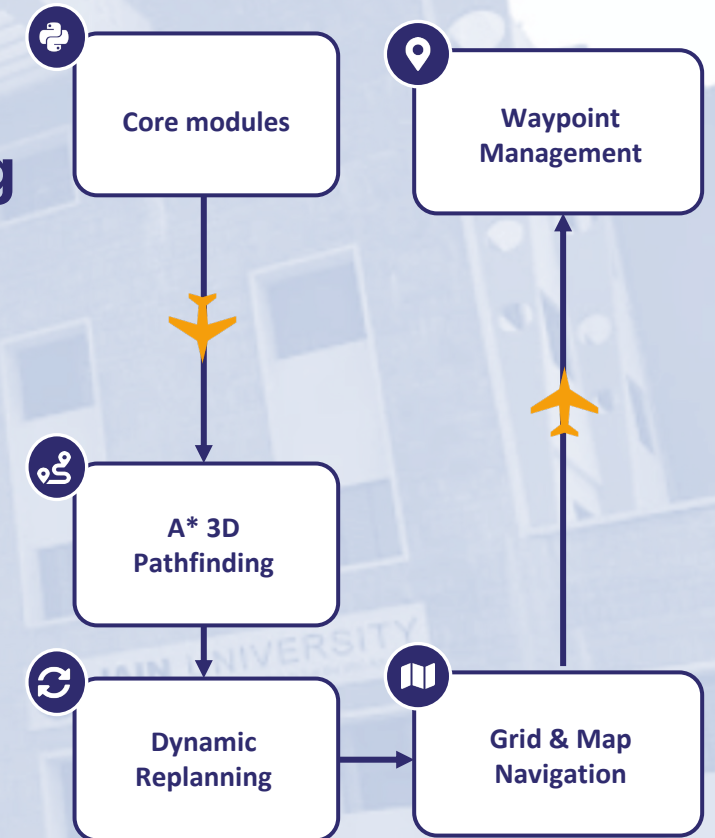
Pygame is used due to its interactive desktop interface, map/grid visualization, controls and mission dashboard



Workflow

II. Core Logic : A* path finding + Dynamic Replanning

- The system uses **A* path-finding** for dynamic 3D route planning around obstacles and hazards
- **3D grid navigation** across X/Y positions and altitude layers
- Supports **dynamic replanning** when hazards are detected and **waypoint navigation** for sequential waypoint and **final-goal routing**.

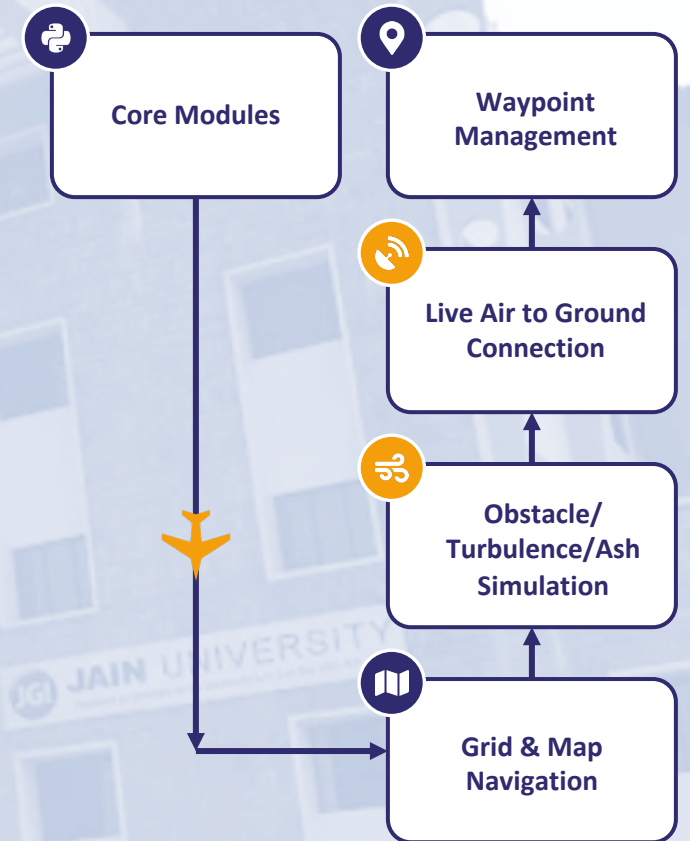


Workflow

III. Simulation & Safety : Environment modeling

The system includes

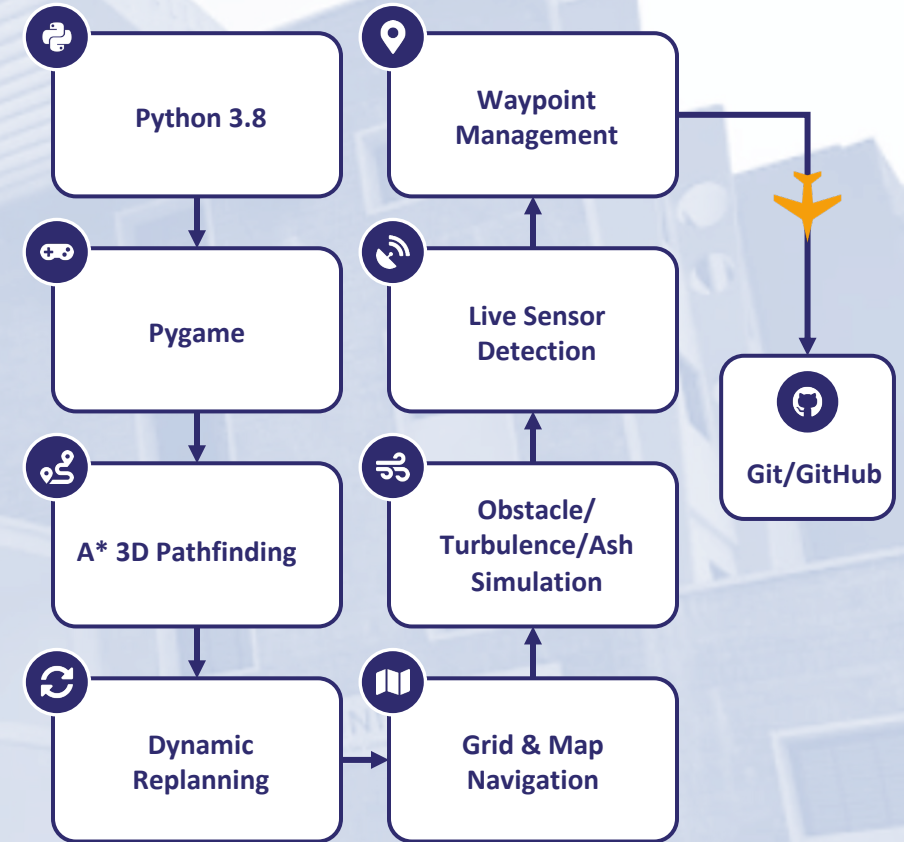
- **obstacle detection**
- **turbulence modeling**
- **ash hazard modeling**
- **live air-to-ground connection**
- **altitude management**



Workflow

IV. Development Tools :

- **GitHub** for publishing
- **Visual Studio Code** for scripting
- **Python Virtual Environment (.venv)** for simulation
- **Pip + requirements.txt** for documentation
- **Claude Sonnet 5.5 and ChatGPT 5.6** for optimization and stress testing



Workflow

Innovation

Utilizing low maintenance software to efficiently and cost effectively create a solution for autonomous vehicles to operate under environmentally degraded conditions by utilizing the drone's connections to the world instead of addition of multiple sensors

Impact

With Janus, low-cost drones and UAVs no longer fly through unnecessary hazardous conditions which help in long-term cost and maintenance savings. It gives users the ability to maintain and safeguard key assets in their inventory

Target Audience / Market

- **Universities: Research Purposes**
- **Schools: Research Purposes**
- **NGOs: Aid**
- **Grassroot humanitarian missions: Aid/Search and rescue**
- **Farmers: Crop monitoring**
- **Rural emergency response staff: Search and Rescue**
- **Serious hobbyists: Experimentation**
- **Low-cost drone manufacturers: Drone Swarms**

Revenue Model

- **A subscription service to a future cloud-based global weather watch which would be hosted on AWS. With this calculated we would price it in the future along with API costs.**
- **An exclusive license for the software to be implemented on majority and eventually all models of low cost of operation UAVs with custom software to plug into any drone.**
- **This will be a Freemium model**

Future Improvements

- **Create a database of global weather information**
- **Pool multiple weather and TFR data points**
- **Make the database accessible to any ground operator of the drone swarm**
- **Use the data to precisely work around crises**
- **Support search and rescue operations**
- **Support other user applications**
- **Help keep drone hardware safe from weather conditions**

THANK YOU